Figure 6.14. A general schematic for how distributed computing works. A master node assigns tasks to worker nodes; the worker nodes perform those tasks and then send their results back to the master node (not shown).

Figure 6.15. The architecture derived from OpenAI's distributed ES paper. Each worker creates a child parameter vector from a parent by adding noise to the parent. Then it evaluates the child's fitness and sends the fitness score to all other agents. Using shared random seeds, each agent can reconstruct the noise vectors used to create the other vectors from the other workers without each having to send an entire vector. Lastly, new parent vectors are created by performing a weighted sum of the child vectors, weighted according to their fitness scores.

Figure 6.16. Figure recreated from the OpenAI "Evolutionary Strategies as a Scalable Alternative to Reinforcement Learning" paper. The figure demonstrates that as more computing resources were added, the time improvement remained constant.

Figure 6.17. The performance of current gradient-based approaches looks like this. In the beginning, there is a seemingly linear trend because the network has not been saturated. But eventually, as more resources are added, we get less and less of a performance boost.

Chapter 7. Distributional DQN: Getting the full story

Figure 7.1. In a randomized control trial of a drug, we study the outcome of some treatment compared to a placebo (a nonactive substance). We want to isolate the effect we are trying to treat, so we take a population with some condition and randomly sort them into two groups: a treatment group and a control group. The treatment group gets the experimental drug we are testing, and the control group gets the placebo. After some time, we can measure the outcome for both groups of patients and see if the treatment group, on average, had a better response than the placebo group.

Figure 7.2. Histogram of the measured blood pressure change for the control and treatment groups in a simulated randomized control trial. The x-axis is the change in blood pressure from the start (before treatment) to after treatment. We want blood pressure to decrease, so